

DATABASE NORMALIZATION PROOF

RDMS - Achieving 3NF Design

1. FIRST NORMAL FORM (1NF)

All tables have atomic attributes (indivisible values). No multi-valued attributes exist.

Evidence:

- User phone numbers stored in single field
- Donor contact stored as single field
- No arrays or composite attributes in any table

2. SECOND NORMAL FORM (2NF)

All non-key attributes are fully dependent on the entire primary key.

Evidence:

- Each table has a single primary key (user_id, beneficiary_id, etc.)
- All attributes in each table depend on the entire primary key
- No partial dependencies exist

3. THIRD NORMAL FORM (3NF)

No transitive dependencies exist. Non-key attributes depend only on primary key.

Evidence:

- Project budget information is stored in Projects table, not in Donations
- Beneficiary demographic data not repeated in Donations table
- All foreign key relationships properly established
- No redundant data across tables

4. RELATIONSHIP INTEGRITY

Primary and foreign keys maintain referential integrity:

- Beneficiaries (PK: beneficiary_id) !•! Aid Distribution (FK: beneficiary_id)
- Donors (PK: donor_id) !•! Donations (FK: donor_id)
- Donations !• Projects through FK relationship
- Users linked to all activity through user_id